

Introduction to Reinforcement Learning

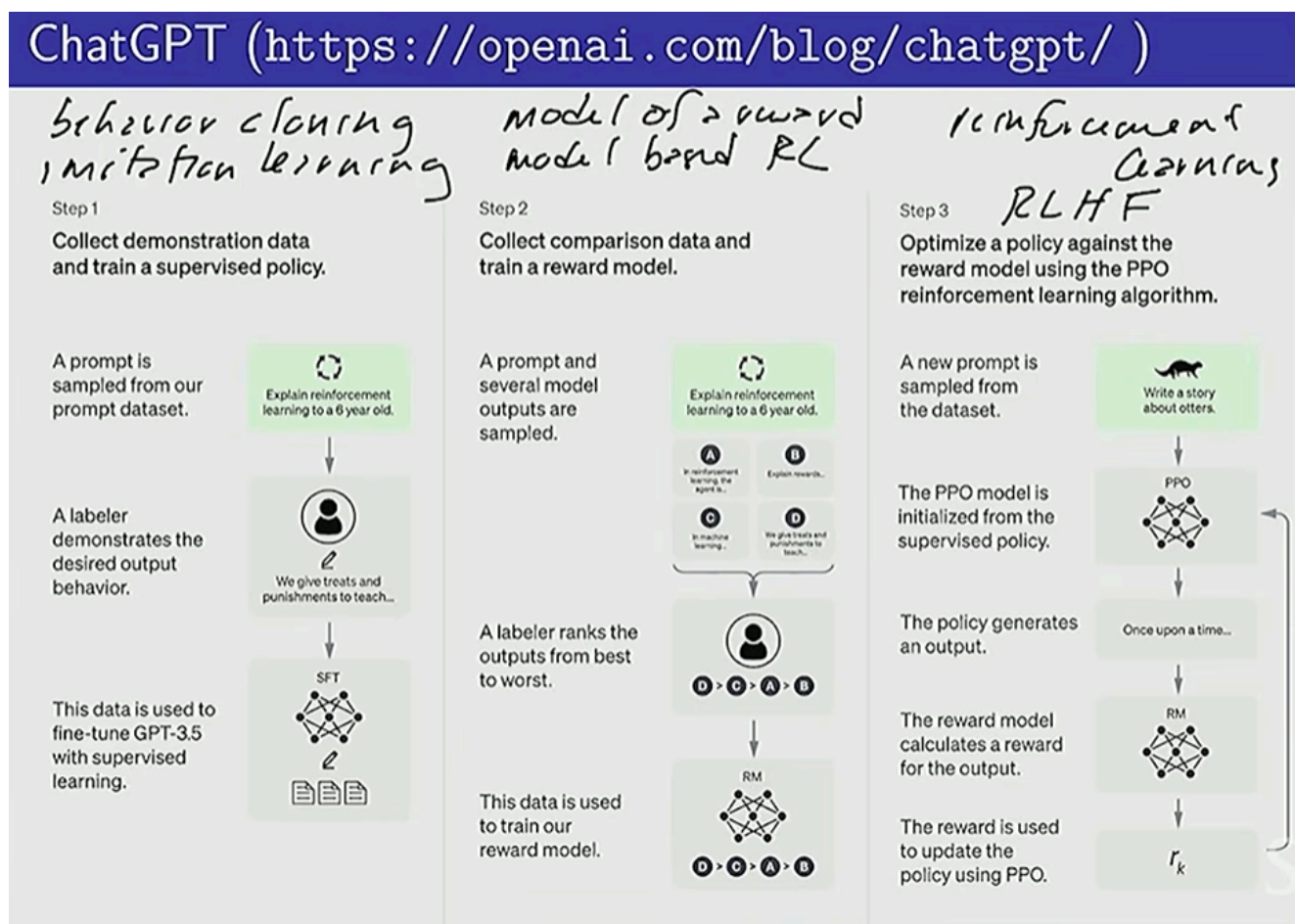
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Tags: ML RL

Intro

How ChatGPT applies RL

1. Behavior cloning (SFT)
2. Model of Reward (model-based reinforcement learning → preference based reward)
3. Reinforcement learning (RLHF: reinforcement learning from Human Feedback)



General Area of RL

- Optimization
- Delayed consequences
- Exploratoion
- Generalization

1. Optimization

Optimization

- Goal is to find an optimal way to make decisions
 - Yielding best outcomes or at least very good outcomes
- Explicit notion of decision utility
- Example: finding minimum distance route between two cities given network of roads

2. Delayed Consequences

Delayed Consequences

- Decisions now can impact things much later...
 - Saving for retirement
 - Finding a key in video game Montezuma's revenge
 - Introduces two challenges
 - When planning: decisions involve reasoning about not just immediate benefit of a decision but also its longer term ramifications
 - When learning: temporal credit assignment is hard (what caused later high or low rewards?)
- Planning → We are not only considering immediate outcome, but multiple things coming together
 - Learning itself → what cause what specifically in that learning
- ### 3. Exploration

Exploration

- Learning about the world by making decisions
 - Agent as scientist
 - Learn to ride a bike by trying (and failing)
- Decisions impact what we learn about
 - Only get a reward for decision made
 - Don't know what would have happened for other decision
 - If we choose to go to Stanford instead of MIT, we will have different later experiences...

4. Generalization

Generalization

- Policy is mapping from past experience to action
- Why not just pre-program a policy?

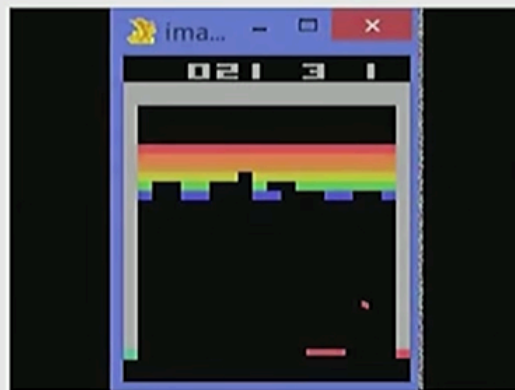


Figure: DeepMind Nature, 2015

- Why pre-programming is not enough
- Because there are soooo many different patterns (or combinations) that is possible, and if these pattern do not have tracable pattern, we cannot write enough policies

Imitation Learning

	AI Planning	SL	UL	RL	IL
Optimization	X			X	X
Learns from experience		X	X	X	X
Generalization	X	X	X	X	X
Delayed Consequences	X			X	X
Exploration				X	

- SL = Supervised learning; UL = Unsupervised learning; RL = Reinforcement Learning; IL = Imitation Learning
- Imitation learning typically assumes input demonstrations of good policies
- IL reduces RL to SL. **For many good reasons, IL is very popular.**

IL here refers to some process like SFT that reduces the RL process back to Supervised Learning.

Something RL is particularly powerful at

1. No examples of desired behaviors
2. Enormous search or optimization problem

- ① No examples of desired behavior: e.g. because the goal is to go beyond human performance or there is no existing data for a task.
- ② Enormous search or optimization problem with delayed outcomes:

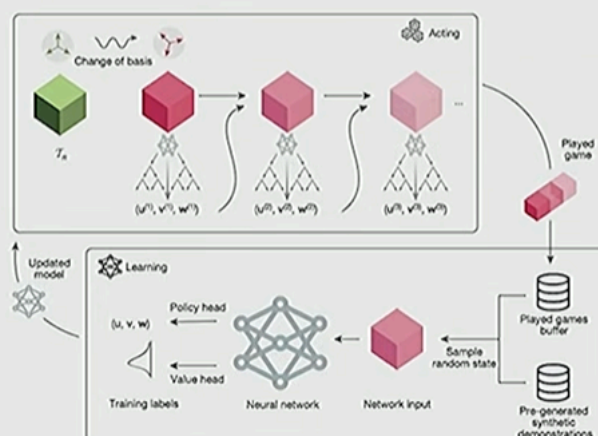


Figure: AlphaTensor. Fawzi et al. 2022

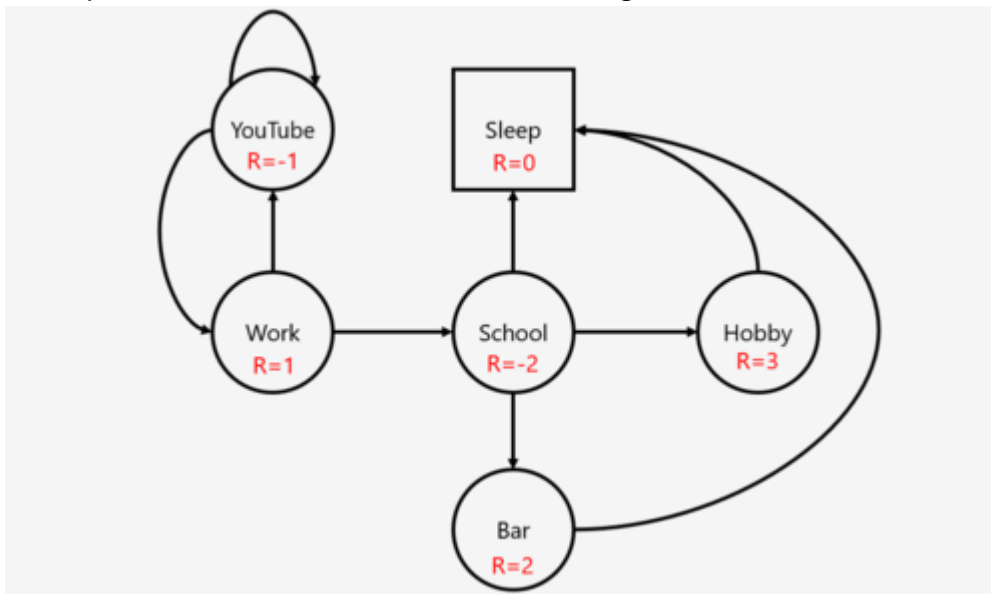
Decision Process

Markov Decision Process

Refresher Exercise: AI Tutor as a Decision Process

- Student initially does not know either addition (easier) nor subtraction (harder)
- AI tutor agent can provide practice problems about addition or subtraction
- AI agent gets rewarded +1 if student gets problem right, -1 if get problem wrong
- Model this as a Decision Process. Define state space, action space, and reward model. What does the dynamics model represent? What would a policy to optimize the expected discounted sum of rewards yield?
- Write down your own answers (5 min) and then discuss in small breakout groups..

example of Markov Decision Process Diagram:



- State:
 1. How good are student at add&subtract problems (0.9, 0.4)
 2. History of all (observed question, reward ...)

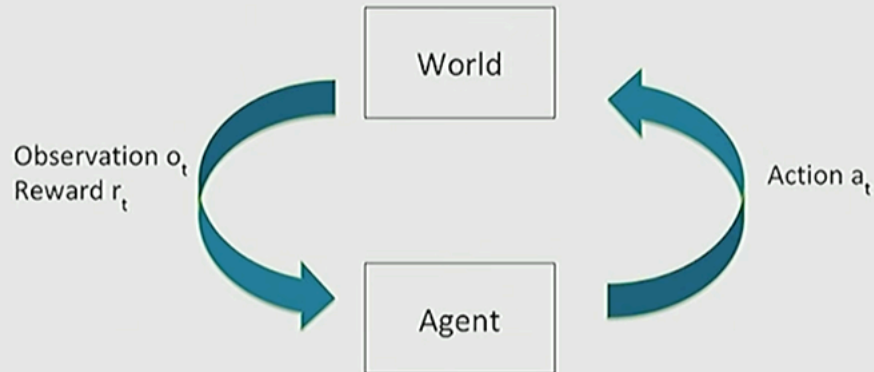
what state means here is the student's knowledge, or the original master of student skill. so we can say that there are 0.9 chance student got add right, 0.4 chance student got subtract right... Something like this.
- Actions: addition question or sub question

- Reward model: +1 if student got it right
- Meaning of dynamics model: agent will only give student easy question, so agent can get lots of award (reward hacking)

Sequential Decision Making

- The statement of history

History: Sequence of Past Observations, Actions & Rewards



- History $h_t = (a_1, o_1, r_1, \dots, a_t, o_t, r_t)$
- Agent chooses action based on history
- State is information assumed to determine what happens next
 - Function of history: $s_t = (h_t)$

eg. use observation on whether you click on amazon advertisement to make action on what type of ads to push to you.

- Markov Assumption

Markov Assumption

- Information state: sufficient statistic of history
- State s_t is Markov if and only if:

$$p(s_{t+1}|s_t, a_t) = p(s_{t+1}|h_t, a_t)$$

- Future is independent of past given present

the second bullet point, Markov here means we can predict the future state **only on the current state**, completely ignoring how we get here.

So for this point, we are saying if we have sufficient statistic of history, we can say that s_t is Markov if and only if this condition works.

what this lead to:

- Simple and often can be satisfied if include some history as part of the state
- In practice often assume most recent observation is sufficient statistic of history: $s_t = o_t$
- State representation has big implications for:
 - Computational complexity
 - Data required
 - Resulting performance

Key is **find the smallest state representation that preserves the Markov property**, i.e., the minimal sufficient statistic of the history. You're compressing without losing the property, not creating the property from scratch.

In this context, the question we are asking here:

1. is state markov? is world partially observable? (POMDP)
2. Are dynamics deterministic or stochastic?
3. Do actions influence only immediate reward (bandits) or reward and next state?

Exmaple: Mars Rover

s_1	s_2	s_3	s_4	s_5	s_6	s_7
+1						+10

Figure: Mars rover image: NASA/JPL-Caltech

- States: Location of rover ($s_1, \dots, s_7$)
- Actions: TryLeft or TryRight
- Rewards:
 - +1 in state s_1
 - +10 in state s_7
 - 0 in all other states

Two things here:

1. Agent's representation of how world changes given agent's action
2. Transition-dynamics model predict next agent state
3. Reward model predicts immediate reward

Policy

Policy

- Policy π determines how the agent chooses actions
- $\pi : S \rightarrow A$, mapping from states to actions
- Deterministic policy:

$$\pi(s) = a$$

- Stochastic policy:

$$\pi(a|s) = Pr(a_t = a | s_t = s)$$

Markov States and Stuff

Markov Process

It is a pair (S, P) :

- S = the finite set of states, here $s_1, \dots, s_7$
- P = a transition model, $P(s_{t+1} = s' \mid s_t = s)$

Markov property: given my current state, the distribution over my next state doesn't care how I got here.

With only 7 states, P is a 7×7 matrix, row i = "if I'm in s_i , here's my distribution over where I land next." Each row sums to 1.

Markov Reward Process (MRP)

An MRP is (S, P, R, γ) — same Markov chain, plus two new things:

$R(s) = \mathbb{E}[r_t \mid s_t = s]$: reward function, expected reward for *being* in a state.

$\gamma \in [0, 1]$: discount factor.

Rover example: $R(s_1) = +1$, $R(s_7) = +10$, $R(\text{everything else}) = 0$.

Return G_t

Once there's a reward at every step, a trajectory produces a sequence of rewards $r_t, r_{t+1}, r_{t+2}, \dots$. The **return** summarizes *how good was this trajectory from time t onward*:

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^{H-1} r_{t+H-1}$$

H = horizon (can be infinite). γ decides how much to discount future rewards (intuitively "a dollar today is worth more than a dollar next year," and mathematically convenient [with $\gamma < 1$, an infinite sum of bounded rewards converges; with $\gamma = 1$ and $H = \infty$ it might not])

Reference

Youtube:

<https://www.youtube.com/watch?v=WsvFL-LjA6U&list=PLoROMvodv4rN4wG6Nk6sNpTEbuOSosZdX>

Textbook:

Sutton and Barton: Chp 1